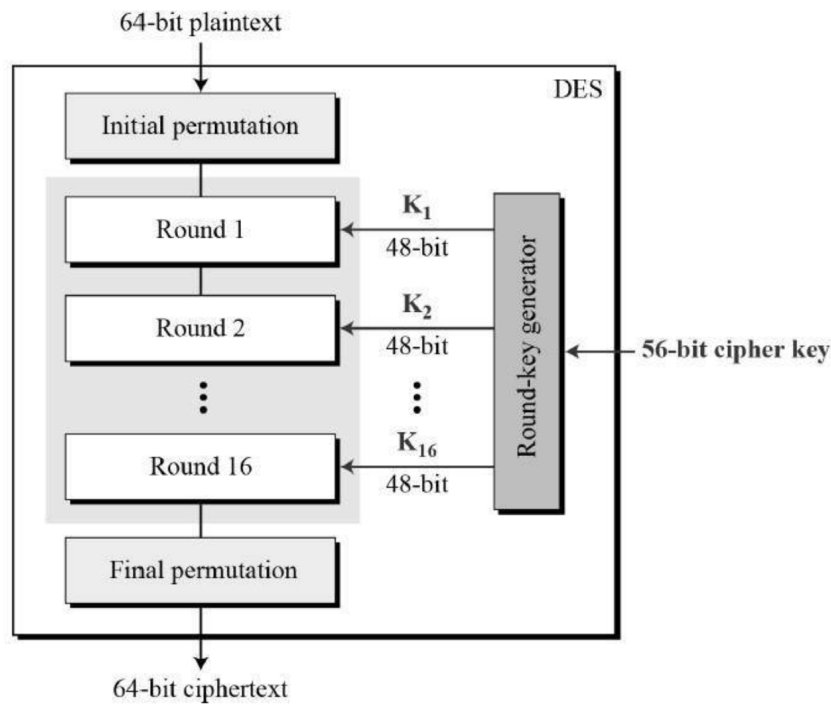


Data Encryption Standard (DES)

Theory:



Initial Permutation (IP):

- The 64-bit plaintext is permuted according to a fixed permutation table.
- The bits are rearranged to make the data suitable for further processing.

Key Generation:

- The 56-bit encryption key is expanded and modified to create 16 subkeys, one for each round.
- Each subkey is 48 bits long, and they are derived through a process of permutation and shifting.

Rounds (16 rounds in total):

- The data is divided into two 32-bit blocks: the left and right halves.
- The right half is expanded to 48 bits using an expansion permutation.
- The expanded right half is XORed with the round's subkey.
- The result goes through substitution using eight S-boxes, which replace 6-bit groups with 4 bits based on fixed tables.
- The outputs from the S-boxes are concatenated and subjected to a fixed permutation.
- The result is XORed with the left half.
- The left and right halves are swapped, and the process is repeated for 16 rounds.

Final Permutation (FP):

- After 16 rounds, the left and right halves are swapped one last time.
- The final permutation is applied to undo the initial permutation and obtain the ciphertext.

Colab File Link: [Lab 2 - DES](#)

Task 1:

DES Encryption:

1. We import the necessary modules from PyCryptodome.
2. The `pad_text` function ensures that the plaintext length is a multiple of 8 bytes to match the DES block size. It appends padding bytes to the plaintext.
3. The `des_encrypt` function performs DES encryption in ECB mode using the provided key.
4. In the main function, we define the plaintext and generate a random DES key.
5. The plaintext is padded, encrypted, and the ciphertext is printed.

DES Decryption:

1. We import the necessary modules from PyCryptodome.
2. The `unpad_text` function removes the padding bytes to retrieve the original plaintext.
3. The `des_decrypt` function decrypts the ciphertext using the provided key.
4. In the main function, replace the ciphertext and key variables with the actual ciphertext and key used for encryption.
5. The ciphertext is decrypted, and the original text is obtained by removing the padding.
6. The decrypted text is printed.

Task 2: Implement double DES Encryption using:

Plaintext = I am Batman

KeyA = b'\x01\xadWR\xeb\x1a\xa2\x86'

KeyB = b'\xf7\xcf\xd6r\xd9\xa1\x141'

Your Output should be

b'\xd2Em\x06-1\xf7\$\x0c\xaaI\xf0\xc3\x00\xbd}'

